Basics about Generative AI

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1 What is Generative AI

1.1 General Definition

Generative AI, sometimes called gen AI, is artificial intelligence (AI) that can create original content and ideas, including conversations, stories, images, videos, music, and software code in response to a user's prompt or request. These systems learn patterns from training data and generate novel outputs that resemble the original data, often powered by architectures like GANs, transformers, diffusion models, and variational autoencoders. Generative AI can be used for various purposes, like chat-bots, media creation, product development, and design.

1.2 How does it work ?

In typical machine learning, practitioners develop artificial intelligence through models that can "learn" from data patterns (find set of rules) without human direction. Instead of following set rules, generative AI studies the basic structure of training data and uses advanced machine learning to generate new content. This lets it make new outputs that match what it learned.

This is what basically gen AI do :

- **It creates new content** like text, images, and videos based on patterns in existing data.
- **It learns complex structures** by analyzing large datasets to understand relationships within the data.
- **It adapts and improves over time** by continuously learning from new data, making its outputs more refined.

1.3 How is Gen AI trained

Generative AI begins with a foundation model, a deep learning model that serves as the basis for multiple different types of generative AI applications. The most common foundation models today are large language models (LLMs), created for text generation applications, but there are also foundation models for image generation, video generation, and sound and music generation as well as multimodal foundation models that can support several kinds content generation.

To create a foundation model, practitioners train a deep learning algorithm on huge volumes of raw, unstructured, unlabeled data e.g., terabytes of data culled from the internet or some other huge data source. During training, the algorithm performs and evaluates millions of 'fill in the blank' exercises, trying to predict the next element in a sequence e.g., the next word in a sentence, the next element in an image, the next command in a line of code and continually adjusting itself to minimize the difference between its predictions and the actual data (or 'correct' result).

The result of this training is a neural network of parameters, encoded representations of the entities, patterns and relationships in the data, that can generate content autonomously in response to inputs, or prompts.

2 Different Types of Gen AI

2.1 Generative adversarial networks (GAN)

GANs consist of two networks—a generator and a discriminator—that compete against each other. The generator creates fake data, while the discriminator evaluates it against real data. The goal is for the generator to produce data indistinguishable from real data.

- Generator : Creates synthetic data based on training.
- Discriminator : Evaluates the data and provides feedback to the generator.

GAN examples

- **Deepfake technology** : GANs can make real videos or images by learning facial movements and expressions. However, these applications raise ethical concerns.
- **Image synthesis** : In industries like fashion or gaming, GANs create high-quality images for product design or character creation.
- **Collaborative art tools** : Certain platforms allow people to create new art pieces or realistic portraits by combining existing images using GAN technology.

2.2 Variational autoencoders (VAE)

VAEs are models that reduce the size of data while keeping its important parts. They then make new data based on that reduced representation. VAEs are commonly used for tasks like : Data compression : Efficiently compress large datasets for easier storage and processing.

- **Image denoising** : Improve the quality of low-resolution or noisy images by generating clearer versions.
- **Medical imaging** : Enhance MRI and CT scan images, offering clearer visuals for diagnosis.

2.3 Transformers

Transformers are a popular architecture in natural language processing and are foundational for language models like GPT-3. These models generate text by predicting the next word in a sentence based on previous context. They operate by using the following methods :

- **Attention mechanisms** : Transformers use self-attention to weigh the importance of each word in a sentence.
- **Language understanding** : Transformers grasp context and relationships between words to generate accurate text.

Transformer applications

- **Content generation** : Write articles, reports, and other forms of creative content with models like GPT-3.
- **Language translation** : Translate text from one language to another with high accuracy.
- **Chatbots** : Provide human-like responses in real time for applications like customer support.

3 Different Content Generated by Gen AI

3.1 Text Generation

Text generation is the process of using Generative AI to produce written content based on a user's prompt. AI models learn patterns in language, including grammar, vocabulary, sentence structure, and meaning, to generate relevant text.

Modern text-generation systems commonly use large language models (LLMs). These models generate text by predicting suitable next tokens based on the preceding context.
How does text generation work ?

1. The user enters a prompt, such as "Explain climate change in simple words."
2. The AI analyzes the prompt and its context.
3. The model generates a sequence of words or tokens that form a response.
4. The generated text is presented to the user, who can review and refine it.

Examples of text generation

- Article writing : Creating articles, blog posts, and reports.
- Story writing : Generating stories, poems, dialogues, and creative content.
- Summarization : Shortening lengthy documents into key points.
- Translation : Translating text between languages.
- Question answering : Answering questions and explaining concepts.
- Email writing : Drafting professional emails and messages.

Tools used

- ChatGPT — generates explanations, stories, summaries, and other text.
- Google Gemini — assists with writing, research, and summarization.
- Claude — supports writing, analysis, and document-based tasks.

Applications Text generation is widely used in education, journalism, customer support, marketing, business communication, and research.

Example prompt : "Write a paragraph about the importance of renewable energy."

Possible output : Renewable energy is important because it helps reduce pollution, decreases dependence on fossil fuels, and supports a more sustainable future.

Limitation : AI-generated text may contain incorrect facts, biased information, or fabricated references. Important information should be verified before use.

3.2 Image Generation

Image generation is the process of creating new visual content using AI. The user can describe an image in words, and the AI generates a visual representation of that description.

Many modern image-generation systems use diffusion models, which learn how images relate to their visual patterns and text descriptions. During generation, these models typically transform a noisy representation into an image that matches the prompt.

How does image generation work ?

1. The user enters a text prompt, such as "A beautiful garden with colorful flowers and butterflies."
2. The AI interprets the objects, colors, style, and scene described.
3. The image-generation model creates an image based on learned patterns.
4. The user can refine the prompt to change the image's appearance.

Examples of image generation

– Digital art : Creating artistic illustrations and paintings.

– Product design : Visualizing products before manufacturing.

– Marketing materials : Creating posters, advertisements, and promotional graphics.

– Educational illustrations : Representing scientific concepts and historical settings.

– Photo-style images : Generating realistic-looking scenes and objects.

– Image editing : Modifying existing images, such as changing backgrounds or styles.

Tools used

– DALL·E — generates images from text prompts.

– Midjourney — creates artistic and stylized images.

– Adobe Firefly — supports AI-powered image creation and editing.

Applications Image generation is useful in graphic design, advertising, entertainment, architecture, education, and product visualization.

Example prompt : "Create a watercolor painting of a small house surrounded by green hills under a blue sky."

The AI can generate a new illustration containing these visual elements in a watercolor style.

Limitation : Generated images may contain distorted objects, inaccurate details, or unusual hands and facial features. Users should also consider copyright, consent, and the risk of misleading or deceptive images.

3.3 Audio Generation

Audio generation is the process of using AI to create sound-based content, such as human-like speech, music, sound effects, and other audio recordings.

AI systems learn patterns in speech, rhythm, melody, pitch, and sound. Depending on the task, they may generate audio directly or convert text into spoken words. **Major types of audio generation**

- Text-to-speech (TTS) : Converts written text into spoken audio.
- Music generation : Creates melodies, songs, and instrumental music.
- Voice generation : Produces synthetic voices or, with appropriate authorization, a voice resembling a particular speaker.
- Sound-effect generation : Creates sounds such as rain, wind, footsteps, or machine noises.
- Speech conversion : Transforms speech into another voice or style.

How does audio generation work ?

1. The user provides text, a music description, or another audio-related prompt.
2. The AI processes the requested content and relevant patterns.
3. The system generates an audio waveform or a sequence of audio units.
4. The resulting audio can be played, downloaded, or edited.

Tools used

- ElevenLabs — provides AI voice generation and text-to-speech capabilities.
- Suno — generates songs and music from prompts.
- Udio — supports AI-assisted music creation.

Applications Audio generation is used in audiobooks, podcasts, virtual assistants, language learning, film production, games, and accessibility tools.

Example prompt : "Generate a calm instrumental music track with piano and soft rain sounds."

An AI audio system can generate a musical track designed to match the requested mood and sound.

Limitation : AI-generated audio may sound unnatural or contain pronunciation errors. Voice cloning can also be misused for impersonation and fraud, so consent and disclosure are important.

3.4 Code Generation

Code generation is the use of Generative AI to write computer programs and software instructions. It helps developers create new code, understand existing code, identify errors, and improve program efficiency.

AI code-generation systems are often trained on programming languages, code examples, documentation, and related technical text. They use the context of a request to suggest code that may solve the stated problem.

How does code generation work ?

– The user describes a programming task, such as "Write a Python program to add two numbers."

– The AI identifies the programming language and requirements.

– It generates code that attempts to meet those requirements.

– The user tests the code, checks the results, and corrects any errors.

Examples of code generation

– Program writing : Creating functions, scripts, and complete program examples.

– Website development : Generating HTML, CSS, and JavaScript.

– Code explanation : Explaining how a program works.

– Debugging assistance : Finding potential errors and suggesting corrections.

– Test generation : Creating test cases to check software behavior.

– Query generation : Writing SQL queries for database operations.

Tools used

– GitHub Copilot — suggests code and assists with programming tasks.

– ChatGPT — generates and explains code in many programming languages.

– Cursor — provides AI-assisted code editing and development features.

Applications Code generation is used in software engineering, web development, data analysis, testing, automation, and programming education.

Limitation : AI-generated code can contain bugs, security vulnerabilities, inefficient logic, or incorrect assumptions. Developers must test and review it before using it in real applications.

3.5 Video Generation

Video generation is the process of creating moving visual content using AI. A system may generate a video from a text description, an image, or existing video footage.

Video-generation models learn patterns involving objects, movement, lighting, and changes between frames. Generating a coherent video is more complex than generating a single image because the content must remain reasonably consistent over time. **How does video generation work ?**

1. The user describes a scene or provides an image.
2. The AI interprets the scene, actions, camera movement, and visual style.
3. The model generates a sequence of frames or video representations.
4. The frames are combined into a video clip, sometimes with generated audio.

Examples of video generation

– Text-to-video : Creating a video from a written description.

– Image-to-video : Animating a still image.

– Animation : Generating animated characters and scenes.

– Advertisement creation : Producing short promotional clips.

– Educational videos : Visualizing scientific or historical concepts.

– Video editing : Modifying or extending existing footage using AI.

Tools used

– Sora — generates video from prompts and supported inputs.

– Runway — offers AI video-generation and editing tools.

– Google Veo — generates video from prompts and supported inputs.

Applications Video generation is useful in filmmaking, marketing, social media, training, entertainment, and educational content creation.

Example prompt : "Create a short video of a train traveling through green mountains on a sunny day."

The system can generate a moving scene representing the train journey.

Limitation : AI-generated videos may contain inconsistent movements, changing object appearances, or unrealistic physical interactions. Synthetic videos can also be used to create misleading content, so authenticity should be considered.

3.6 Challenges and responsible use

Although Generative AI is useful, it also presents important challenges :

– Accuracy : AI may produce false or misleading information.

– Bias : Generated content can reflect biases in the data used to train the model.

– Copyright : Some generated outputs may raise questions about ownership or the use of protected material.

– Privacy : Users should avoid entering confidential or sensitive information into systems without understanding their data policies.

– Misuse : Generated images, voices, videos, and text can be used for deception or impersonation.

– Human supervision : Generated outputs should be checked for quality, safety, and suitability.

4 Generative AI model architectures and how they have evolved

Truly generative AI models deep learning models that can autonomously create content on demand have evolved over the last dozen years or so. The milestone model architectures during that period include

Variational autoencoders (VAEs), which drove breakthroughs in image recognition, natural language processing and anomaly detection.

Generative adversarial networks (GANs) and diffusion models, which improved the accuracy of previous applications and enabled some of the first AI solutions for photo-realistic image generation.

Transformers, the deep learning model architecture behind the foremost foundation models and generative AI solutions today.

4.1 Variational autoencoders (VAEs)

An autoencoder is a deep learning model comprising two connected neural networks : One that encodes (or compresses) a huge amount of unstructured, unlabeled training data into parameters, and another that decodes those parameters to reconstruct the content. Technically, autoencoders can generate new content, but they're more useful for compressing data for storage or transfer, and decompressing it for use, than they are for high-quality content generation.

Introduced in 2013, variational autoencoders (VAEs) can encode data like an autoencoder, but decode multiple new variations of the content. By training a VAE to generate variations toward a particular goal, it can 'zero in' on more accurate, higher-fidelity content over time. Early VAE applications included anomaly detection (e.g., medical image analysis) and natural language generation.

4.2 Generative adversarial networks (GANs)

GANs, introduced in 2014, also comprise two neural networks : A generator, which generates new content, and a discriminator, which evaluates the accuracy and quality the generated data. These adversarial algorithms encourages the model to generate increasingly high-quality outpits.

GANs are commonly used for image and video generation, but can generate high-quality, realistic content across various domains. They've proven particularly successful at tasks as style transfer (altering the style of an image from, say, a photo to a pencil sketch) and data augmentation (creating new, synthetic data to increase the size and diversity of a training data set).

4.3 Diffusion models

Also introduced in 2014, diffusion models work by first adding noise to the training data until it's random and unrecognizable, and then training the algorithm to iteratively diffuse the noise to reveal a desired output.

Diffusion models take more time to train than VAEs or GANs, but ultimately offer finer-grained control over output, particularly for high-quality image generation tool. DALL-E, Open AI's image-generation tool, is driven by a diffusion model.

4.4 Transformers

First documented in a 2017 paper published by Ashish Vaswani and others, transformers evolve the encoder-decoder paradigm to enable a big step forward in the way foundation models are trained, and in the quality and range of content they can produce. These models are at the core of most of today's headline-making generative AI tools, including ChatGPT and GPT-4, Copilot, BERT, Bard, and Midjourney to name a few.
Transformers use a concept called attention, determining and focusing on what's most important about data within a sequence to ;
process entire sequences of data e.g., sentences instead of individual words simultaneously ;

capture the context of the data within the sequence ;

encode the training data into embeddings (also called hyperparameters) that represent the data and its context.

In addition to enabling faster training, transformers excel at natural language processing (NLP) and natural language understanding (NLU), and can generate longer sequences of data e.g., not just answers to questions, but poems, articles or papers with greater accuracy and higher quality than other deep generative AI models. Transformer models can also be trained or tuned to use tools e.g., a spreadsheet application, HTML, a drawing program to output content in a particular format.

5 What are the limitations of generative AI ?

Despite their advancements, generative AI systems can sometimes produce inaccurate or misleading information. They rely on patterns and data they were trained on and can reflect biases or inaccuracies inherent in that data. Other concerns related to training data include

5.1 Security

Data privacy and security concerns arise if proprietary data is used to customize generative AI models. Efforts must be made to ensure that the generative AI tools generate responses that limit unauthorized access to proprietary data. Security concerns also arise if there is a lack of accountability and transparency in how AI models make decisions.

5.2 Creativity

While generative AI can produce creative content, it often lacks true originality. The creativity of AI is bounded by the data it has been trained on, leading to outputs that may feel repetitive or derivative. Human creativity, which involves a deeper understanding and emotional resonance, remains challenging for AI to replicate fully.

5.3 Cost

Training and running generative AI models require substantial computational resources. Cloud-based generative AI models are more accessible and affordable than trying to build new models from scratch.

5.4 Explainability

Due to their complex and opaque nature, generative AI models are often considered black boxes. Understanding how these models arrive at specific outputs is challenging. Improving interpretability and transparency is essential to increase trust and adoption.